

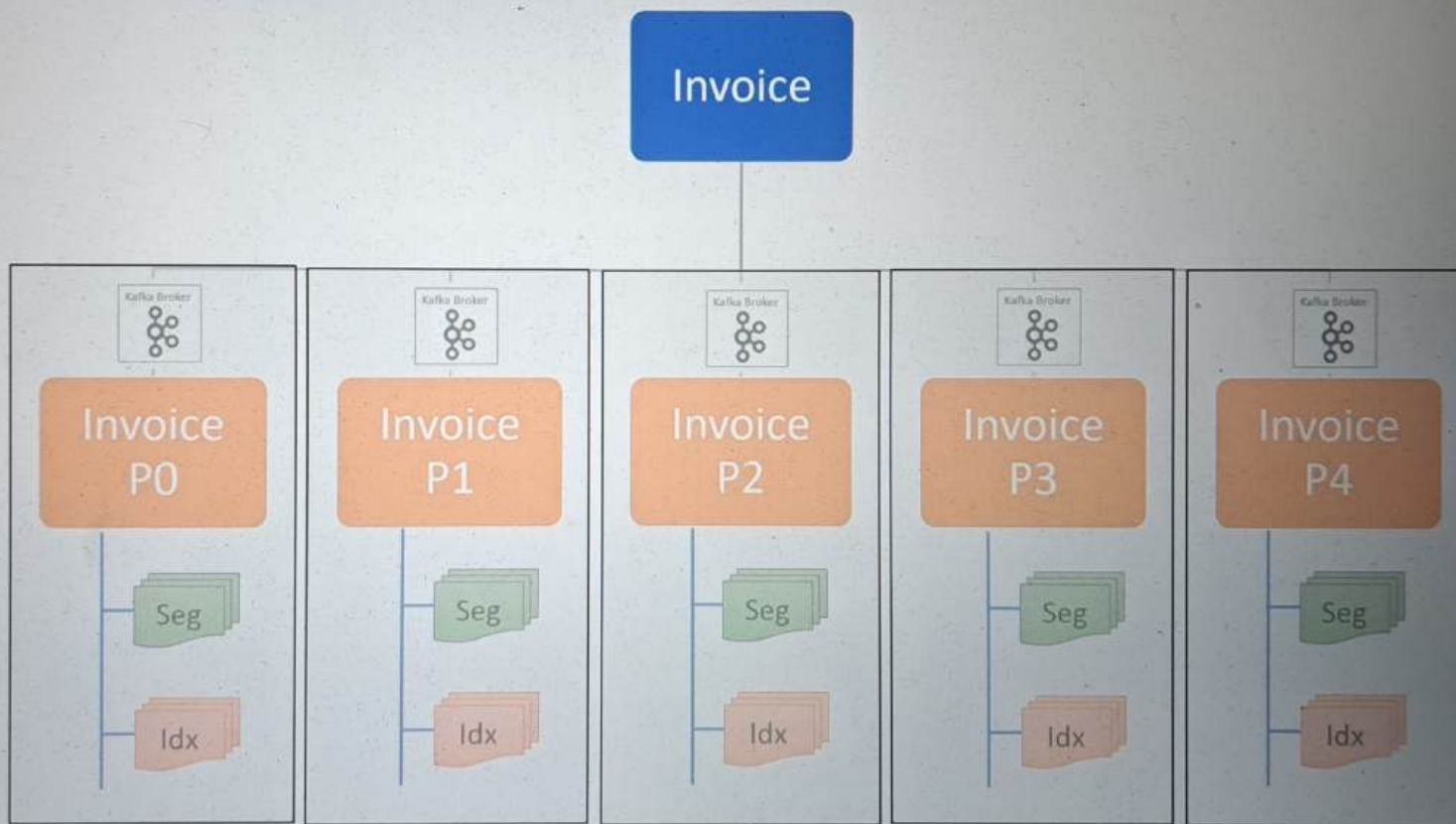
Kafka Architecture

1. Partitions – Log Files

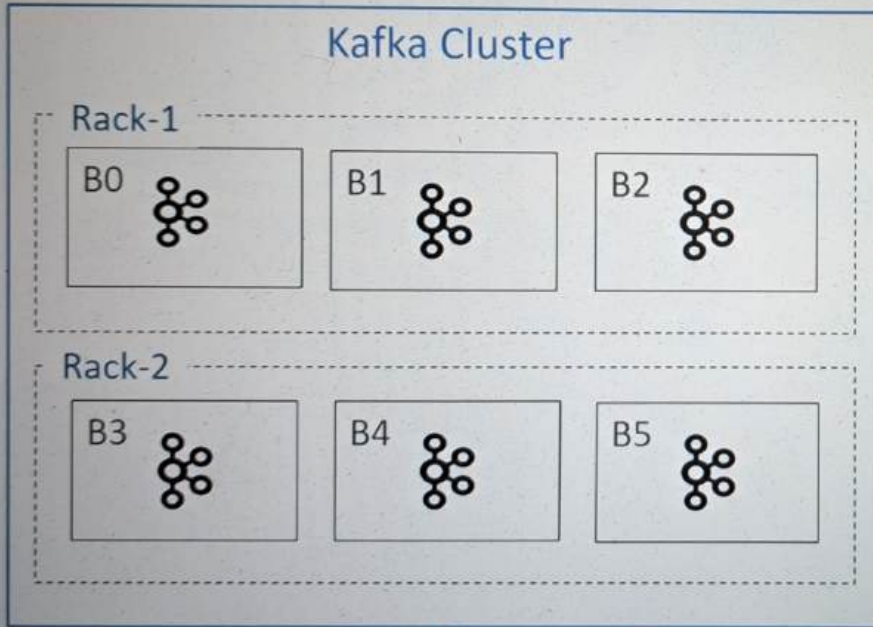
→ 2. Cluster Formation

Kafka Fault Tolerance?

Kafka Architecture



Partition Assignment

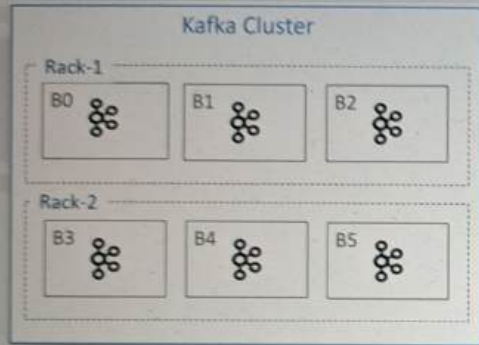


Topic - Invoice

1. Partitions -> 10
2. Replication Factor -> 3

Total Replicas = 30

1. Even Distribution
2. Fault Tolerance.



Partition Assignment

1. Ordered List of Brokers
- ➔ 2. Leader and Followers Assignment

Kafka Cluster

Rack-1



Rack-2



Partition Assignment

R1-B0 

R2-B3 

R1-B1 

R2-B4 

R1-B2 

R2-B5 



Partition Assignment

P0	P0	P0
P1	P1	P1
P2	P2	P2
P3	P3	P3
P4	P4	P4
P5	P5	P5
P6	P6	P6
P7	P7	P7
P8	P8	P8
P9	P9	P9

Brokers

Leaders

Followers

Followers

R1-B0 

R2-B3 

R1-B1 

R2-B4 

R1-B2 

R2-B5 

Partition Assignment

Brokers	Leaders		Followers		Followers	
R1-B0 	P0	P6		P5		P4
R2-B3 	P1	P7	P0	P6		P5
R1-B1 	P2	P8	P1	P7	P0	P6
R2-B4 	P3	P9	P2	P8	P1	P7
R1-B2 	P4		P3	P9	P2	P8
R2-B5 	P5		P4		P3	P9

Partition Assignment

Brokers	Leaders		Followers		Followers	
R1-B0 	P0	P6		P5		P4
R2-B3 	P1	P7	P0	P6		P5
R1-B1 	P2	P8	P1	P7	P0	P6
R2-B4 	P3	P9	P2	P8	P1	P7
R1-B2 	P4		P3	P9	P2	P8
R2-B5 	P5		P4		P3	P9